

MONAURAL CONTINUOUS-RADIUS REGIONAL SPEECH EXTRACTION WITH CROSS-RADIUS CONSISTENCY LEARNING

Author(s) Name(s)

Author Affiliation(s)

ABSTRACT

Source-to-microphone distance provides a speaker-independent cue for selective speech extraction without requiring prior enrollment. We propose a monaural continuous-radius regional extraction framework that preserves speech within a queried region while suppressing speech outside it. The query radius is represented as a continuous scalar and injected into a time-frequency extraction network, enabling arbitrary-radius queries with a single model. We exploit the nested structure of target sets across radii and introduce cross-radius consistency learning to stabilize outputs when the target set remains unchanged. Experiments show that continuous radius conditioning substantially improves selective extraction. With consistency learning, the model achieves 29.25 dB SI-SDR for full-target preservation, 4.40 dB SI-SDR_i for partial-target extraction, 62.01 dB attenuation for empty-target rejection, and 0.72% radius consistency error. The model generalizes to multi-speaker mixtures and maintains robustness under noisy conditions.

Index Terms— regional speech extraction, distance conditioning, continuous radius, consistency learning

1. INTRODUCTION

Selective speech extraction is important for speech communication and listening applications in multi-talker environments. Target speech extraction (TSE) [1, 2] aims to recover speech of interest from an acoustic mixture according to a user-specified cue, with speaker identity being one of the most commonly adopted forms of conditioning [3, 4]. However, speaker-conditioned TSE typically requires prior enrollment or other target-specific information, limiting its flexibility when the desired speaker is unknown or changes dynamically. Spatial cues provide an alternative by specifying target speech according to location rather than speaker identity. Among them, source-to-microphone distance provides a simple and speaker-independent cue for selecting speech according to its proximity to the recording device.

Recent studies have demonstrated the utility of source-to-microphone distance for monaural speech separation and extraction. Patterson *et al.* [5] introduced distance-based

sound separation, which separates near and far sources according to a predefined distance threshold. Lin *et al.* [6] further incorporated distance and speaker information into monaural TSE. More recently, Shi *et al.* [7] proposed monaural TSE conditioned on a query distance to extract speech within a local range around that distance. Their subsequent work [8] additionally incorporated room-related information to improve robustness across acoustic environments, highlighting the dependence of monaural distance cues on room acoustics. These studies establish distance as a viable cue for monaural speech extraction and provide a basis for extending distance-based target selection toward regional speech extraction.

Regional speech extraction has also been explored using multichannel spatial information. Gu and Luo [9] proposed ReZero, which extracts sounds from user-defined spatial regions using microphone-array observations. Chen *et al.* [10] developed programmable sound bubbles for hearable devices, preserving sounds within a specified radius while suppressing those outside the region. More recently, regional speech separation methods further exploited directional and distance cues from microphone arrays [11]. These studies demonstrate the feasibility of flexible regional speech extraction; however, their spatial selectivity relies on multichannel observations. Therefore, continuous-radius regional speech extraction using only a single microphone remains largely unexplored.

In this work, we investigate monaural continuous-radius regional speech extraction, in which a monaural mixture is conditioned on a query radius to preserve all speech sources within the specified region while suppressing those outside it. As illustrated in Fig. 1, we represent the query radius as a continuous scalar and inject its embedding into the extraction network, enabling a single model to operate over continuously adjustable regions. Furthermore, we exploit the nested structure of radius-defined targets and introduce a cross-radius consistency objective to encourage stable predictions when different query radii correspond to the same target-source set. Experiments under diverse source configurations evaluate regional selectivity, continuous-radius control, and cross-radius consistency, demonstrating the effectiveness of the proposed framework.

Thanks to XYZ agency for funding.

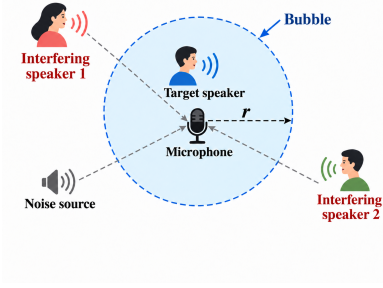


Fig. 1. Task overview.

2. METHOD

2.1. Problem formulation

Consider a monaural acoustic scene containing N speakers. The observed mixture is formulated as

$$x(t) = \sum_{i=1}^N s_i(t) + n(t), \quad (1)$$

where $x(t)$ denotes the monaural mixture at time t , $s_i(t)$ denotes the microphone-received signal from the i th speaker, and $n(t)$ denotes additive noise. Let d_i denote the distance between the i th speaker and the microphone. Given a query radius r , we define the region centered at the microphone with radius r as the target region and the sum of all speaker signals within this region as the desired signal:

$$y_r(t) = \sum_{i: d_i < r} s_i(t). \quad (2)$$

Here, $y_r(t)$ denotes the desired speech signal associated with the query radius r . If no speaker lies inside the region, $y_r(t)$ is a zero signal. Following cue-conditioned target speech extraction formulations [3, 4, 6, 7], our objective is to learn a monaural distance-conditioned speech extraction model f_θ , parameterized by θ , that estimates the desired signal from the mixture x and the query radius r :

$$\hat{y}_r = f_\theta(x, r). \quad (3)$$

Instead of treating distance as a set of discrete classes, we formulate the query radius as a continuous variable, $r \in [0, R_{\max}]$, where $R_{\max}$ denotes the maximum query radius supported by the system. This formulation allows a single model to dynamically adjust the spatial extent of the region according to r without restricting the query to a predefined set of discrete distances. Furthermore, for two query radii satisfying $0 \leq r_1 < r_2 \leq R_{\max}$, we define the corresponding target-speaker set as $\mathcal{S}(r) = \{i \mid d_i < r\}$, which naturally satisfies $\mathcal{S}(r_1) \subseteq \mathcal{S}(r_2)$. This inclusion property indicates that every target speaker within the smaller region must also belong to the larger one. Consequently, the target signals associated with different query radii exhibit a nested structure.

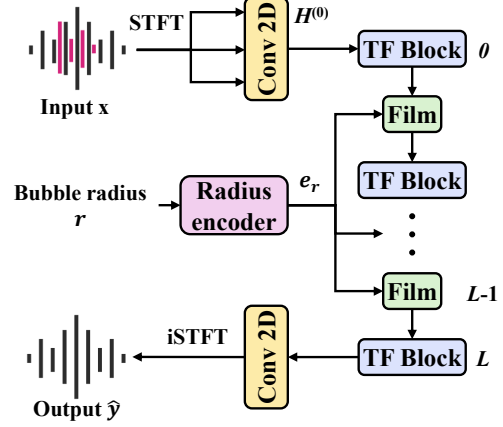


Fig. 2. Overview of the continuous radius-conditioned extraction network.

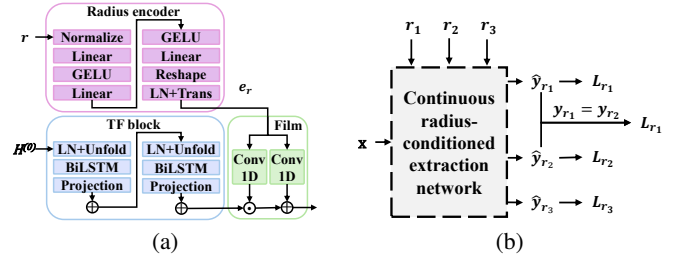


Fig. 3. Illustrations of (a) radius-conditioned GridNet processing and (b) cross-radius consistency learning.

This nested structure further reveals a local invariance property: when the target-speaker set remains unchanged ($\mathcal{S}(r_1) = \mathcal{S}(r_2)$), the desired signal should also remain identical ($y_{r_1}(t) = y_{r_2}(t)$). Conversely, when increasing the radius introduces additional speakers into the region, the desired signal changes accordingly by incorporating the newly included sources. Therefore, for a fixed acoustic scene, the desired signal exhibits a piecewise-invariant structure with respect to the query radius: it remains stable over radius intervals that do not cross any speaker distance while changing only when the region boundary passes one or more speakers. This physically motivated property provides the basis for the cross-radius consistency learning introduced in Sec. 2.3.

2.2. Continuous radius-conditioned extraction network

Fig. 2 illustrates the proposed continuous radius-conditioned extraction network. We employ the lightweight TF-GridNet [12] backbone for monaural speech extraction. Given a mixture x , its magnitude-normalized complex spectrum $X \in \mathbb{C}^{F \times T}$ is obtained by a short-time Fourier transform (STFT). The real, imaginary, and log-magnitude components are concatenated along the feature dimension and projected by an input convolution into $H^{(0)} \in \mathbb{R}^{C \times T \times F}$, where C , T , and F denote the

number of feature channels, time frames, and frequency bins, respectively.

The backbone consists of L sequentially stacked GridNet blocks. As shown in Fig. 3(a), given the input $H^{(l)}$ to the l th block, bidirectional sequence modeling is first performed along the frequency axis:

$$U^{(l)} = H^{(l)} + P_l^f(\text{BiLSTM}_l^f(\text{LN}_l^f(H^{(l)}))). \quad (4)$$

A second stage is then performed along the temporal axis:

$$V^{(l)} = U^{(l)} + P_l^t(\text{BiLSTM}_l^t(\text{LN}_l^t(U^{(l)}))). \quad (5)$$

Here, $\text{LN}(\cdot)$, $\text{BiLSTM}(\cdot)$, and $P(\cdot)$ denote feature normalization, bidirectional LSTM, and linear projection respectively, while superscripts f and t indicate the frequency and temporal modeling paths. The two residual paths successively capture spectral and temporal dependencies while preserving the time-frequency feature dimensionality.

To support continuously adjustable regions, the query radius r is directly treated as a continuous scalar and encoded by a three-layer MLP into a frequency-dependent representation e_r . As illustrated in Fig. 3(a), radius information is then injected between consecutive GridNet blocks through feature-wise linear modulation (FiLM) [13]:

$$H^{(l)} = \Gamma_{l-1}(e_r) \odot V^{(l-1)} + B_{l-1}(e_r), \quad l = 1, \dots, L-1, \quad (6)$$

where $\Gamma_l(\cdot)$ and $B_l(\cdot)$ generate scaling and bias parameters matched to the backbone features, and $\odot$ denotes element-wise multiplication. This multi-stage conditioning allows the query radius to continuously regulate the intermediate time-frequency representations and enables arbitrary radius queries within the training range. Finally, the output of the last GridNet block is decoded into the real and imaginary components of the target complex spectrum, rescaled using the front-end normalization factor, and reconstructed into the waveform $\hat{y}_r$ through inverse STFT (ISTFT).

2.3. Cross-radius consistency learning

To explicitly exploit the nested structure across regions, we sample Q query radii $r_1 < r_2 < \dots < r_Q$ from the continuous interval $[0, R_{\max}]$ for the same mixture x , where $Q \geq 2$. Following the local invariance established in Sec. 2.1, consistency is imposed only on adjacent radius pairs whose target-speaker sets remain unchanged:

$$\mathcal{P} = \{(r_q, r_{q+1}) \mid \mathcal{S}(r_q) = \mathcal{S}(r_{q+1})\}, \quad q = 1, \dots, Q-1. \quad (7)$$

If increasing the radius introduces additional speakers into the region, the desired signal should change accordingly, and the corresponding pair is excluded from the consistency constraint to avoid oversmoothing across speaker boundaries.

As illustrated in Fig. 3(b), for each valid pair $(r_a, r_b) \in \mathcal{P}$, we minimize the waveform discrepancy between the two

predictions normalized by the energy of their common target:

$$\mathcal{L}_{\text{con}} = \frac{1}{|\mathcal{P}|} \sum_{(r_a, r_b) \in \mathcal{P}} \frac{\|\hat{y}_{r_a} - \hat{y}_{r_b}\|_1}{\text{RMS}(y_{r_a}) + \epsilon}. \quad (8)$$

Here, $|\mathcal{P}|$ denotes the number of valid radius pairs, and $\text{RMS}(\cdot)$ denotes the root-mean-square amplitude. The normalization reduces the influence of target-energy variation, while pairs with an empty common target are excluded from this term.

2.4. Training objective

For each query radius r_q , the training objective is determined by whether the corresponding target signal is empty. For notational brevity, let $\hat{y}_q = \hat{y}_{r_q}$, $y_q = y_{r_q}$, and $e_q = \hat{y}_q - y_q$. For non-empty targets, we use the SNR-based reconstruction loss $\ell_{\text{SNR}}(\hat{y}_q, y_q) = 10 \log_{10}(1 + \text{MSE}(e_q)/(\text{MSE}(y_q) + \epsilon))$:

$$\mathcal{L}_q = \begin{cases} \ell_{\text{SNR}}(\hat{y}_q, y_q), & y_q \neq 0, \\ \alpha \cdot \text{mean}(|\hat{y}_q|), & y_q = 0, \end{cases} \quad (9)$$

where $\text{MSE}(z) = \text{mean}(z^2)$, ϵ is a numerical stability term, and α denotes the zero-target suppression weight. The ℓ_1 loss encourages a near-zero output for empty targets. The primary extraction loss $\mathcal{L}_{\text{ext}}$ is obtained by averaging $\mathcal{L}_q$ over the Q query radii. The overall training objective is

$$\mathcal{L} = \mathcal{L}_{\text{ext}} + \lambda_{\text{con}} \mathcal{L}_{\text{con}}, \quad (10)$$

where $\mathcal{L}_{\text{con}}$ denotes the cross-radius consistency loss, and λ_{con} controls its contribution.

3. EXPERIMENTS

3.1. Experimental setup

Data. All experiments use simulated monaural data. We generate RIRs for 1,000 random rooms using `gpuRIR` [14] and split them into 800/100/100 rooms for training, validation, and testing. All signals are processed at 16 kHz with a duration of 4.5 s. Source-to-microphone distances are grouped into five intervals, $[0, 1)$, $[1, 2)$, $[2, 3)$, $[3, 4)$, and $[4, +\infty)$ m, yielding one no-speaker, five single-speaker, and 15 two-speaker profiles. During training, noise is added with a probability of 0.5 at 0–15 dB SNR. Continuous query radii are sampled within the distance intervals, and the target is defined as the sum of speakers inside the query radius.

Implementation details. The FFT length and hop size of the STFT are set to 192 and 128 samples, respectively. The model uses $L = 6$ GridNet blocks, a backbone feature dimension of 32, and a bidirectional LSTM hidden dimension of 64, resulting in approximately 0.66 M trainable parameters. We train the model with the Adam optimizer using an initial learning rate of 6×10^{-4} , a warm-up of 2,000 optimization

Table 1. Comparison and ablation study.

Radius conditioning	Consistency	Full SI-SDR ↑	Partial SI-SDRi ↑	Empty Atten. ↑	RCE ↓
Discrete	×	30.43	1.82	49.56	2.15%
Discrete	✓	28.75	3.15	57.70	0.81%
Continuous	×	27.04	3.08	51.13	0.85%
Continuous	✓	29.25	4.40	62.01	0.72%

steps, and a ReduceLRonPlateau scheduler. The batch size is 1, the maximum number of training epochs is 50, and the cross-radius consistency weight λ_{con} is set to 0.1.

Evaluation metrics. According to the target-speaker set specified by the query radius, we use SI-SDR [15] for *full-target* cases to measure speech preservation, SI-SDR improvement (SI-SDRi) for *partial-target* cases to evaluate selective extraction, and input-to-output attenuation (Atten.) for *empty-target* cases to quantify suppression of sources outside the region. We further define the Radius Consistency Error (RCE) over adjacent radius pairs $\mathcal{P}$ satisfying $\mathcal{S}(r_a) = \mathcal{S}(r_b)$:

$$\text{RCE} = \frac{1}{|\mathcal{P}|} \sum_{(r_a, r_b) \in \mathcal{P}} \frac{\|\hat{y}_{r_a} - \hat{y}_{r_b}\|_1}{T(\text{RMS}(x) + \epsilon)}. \quad (11)$$

A lower RCE indicates better cross-radius consistency.

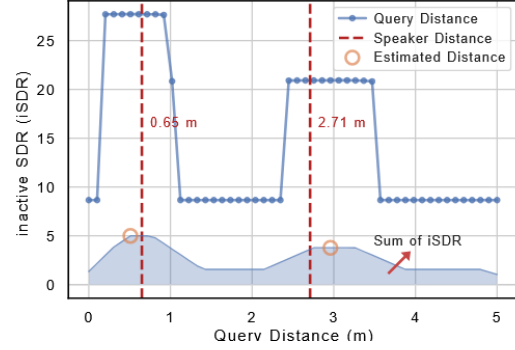
3.2. Ablation Study

We compare discrete and continuous radius conditioning in Table 1. Without consistency regularization, continuous conditioning substantially improves partial-target extraction and reduces RCE, indicating more stable behavior across radius queries. With consistency enabled, the continuous model achieves further improvements across all metrics.

We observe that the consistency objective consistently improves cross-radius stability under both discrete and continuous conditioning while also benefiting partial-target extraction and empty-target rejection. Overall, the combination of continuous conditioning and consistency learning achieves the best SI-SDRi, attenuation, and RCE while maintaining high full-target fidelity.

3.3. Continuous-Radius Analysis

We show the model behavior under continuously varying query radii in Fig. 4. The output remains largely stable when the radius varies between adjacent speaker distances, while a clear transition occurs once the radius crosses the physical distance of a speaker, incorporating the newly included target accordingly. This behavior agrees with the piecewise-invariant target structure defined in Sec. 2.1, indicating that the model supports continuously adjustable region selection rather than merely learning a set of discrete query modes. The low RCE reported in Table 1 further confirms stable cross-radius predictions when the target-speaker set remains unchanged.

**Fig. 4.** Continuous-radius analysis of the estimated output.**Table 2.** Evaluation under challenging conditions.

Condition	Full SI-SDR ↑	Partial SI-SDRi ↑	Empty Atten. ↑	RCE ↓
2-spkr, Clean	29.25	4.40	62.01	0.72%
2-spkr, 10 dB	29.05	4.23	61.76	0.79%
2-spkr, 5 dB	28.38	3.92	60.47	0.82%
3-spkr, Clean	26.17	3.19	59.28	0.88%
3-spkr, 10 dB	24.98	3.10	52.67	0.86%

3.4. Evaluation under Challenging Conditions

To further evaluate model performance under more challenging conditions, we consider scenarios with increased speaker number and additive noise. As shown in Table 2, performance decreases when the number of speakers increases from two to three; however, the model retains 3.19 dB partial-target SI-SDRi, 59.28 dB empty-target attenuation, and an RCE below 0.9%. This result indicates that the model generalizes to more complex nested target sets while maintaining stable predictions across radius queries.

For noisy evaluation, we test the model under clean, 10 dB, and 5 dB input SNR conditions. As the SNR decreases, SI-SDR, SI-SDRi, and attenuation under the two-speaker condition gradually degrade; however, RCE increases only from 0.72% to 0.82%, indicating limited impact on cross-radius stability. Under the three-speaker 10 dB condition, the model still maintains 24.98 dB SI-SDR, 3.10 dB SI-SDRi, 52.67 dB attenuation, and 0.86% RCE, demonstrating robust radius-conditioned extraction under both noisy and multi-speaker conditions.

4. CONCLUSION

We investigate continuous-radius regional extraction from monaural mixtures and propose a radius-conditioned speech extraction framework. We model the query radius as a continuous variable and inject it into the time-frequency backbone, while a cross-radius consistency objective exploits the piecewise-invariant structure of targets across differ-

ent radii to improve prediction stability. Experiments show that continuous radius conditioning enables arbitrary-radius queries while maintaining effective region-based extraction, and that consistency learning further improves selective extraction, empty-region rejection, and cross-radius stability. Continuous-radius analysis confirms stable outputs between speaker boundaries and clear transitions when the query radius crosses a source distance. The model also generalizes to increased speaker numbers and additive noise. Future work will investigate robustness and generalization in real acoustic environments.

5. REFERENCES

- [1] Yi Luo and Nima Mesgarani, “Conv-tasnet: Surpassing ideal time-frequency magnitude masking for speech separation,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 27, no. 8, pp. 1256–1266, Aug. 2019.
- [2] Cem Subakan, Mirco Ravanelli, Samuele Cornell, Mirko Bronzi, and Jianyuan Zhong, “Attention is all you need in speech separation,” in *2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. June 2021, pp. 21–25, IEEE.
- [3] Kateřina Žmolíková, Marc Delcroix, Keisuke Kinoshita, Takuya Higuchi, Atsunori Ogawa, and Tomohiro Nakatani, “Speaker-aware neural network based beamformer for speaker extraction in speech mixtures,” in *Interspeech 2017*. Aug. 2017, pp. 2655–2659, ISCA.
- [4] Quan Wang, Hannah Muckenhirn, Kevin Wilson, Prashant Sridhar, Zelin Wu, John R. Hershey, Rif A. Saurous, Ron J. Weiss, Ye Jia, and Ignacio Lopez Moreno, “Voicefilter: Targeted voice separation by speaker-conditioned spectrogram masking,” in *Interspeech 2019*. Sept. 2019, pp. 2728–2732, ISCA.
- [5] Katharine Patterson, Kevin Wilson, Scott Wisdom, and John R. Hershey, “Distance-based sound separation,” in *Interspeech 2022*. Sept. 2022, pp. 901–905, ISCA.
- [6] Jiuxin Lin, Peng Wang, Heinrich Dinkel, Jun Chen, Zhiyong Wu, Zhiyong Yan, Yongqing Wang, Junbo Zhang, and Yujun Wang, “Focus on the sound around you: Monaural target speaker extraction via distance and speaker information,” in *Interspeech 2023*. Aug. 2023, pp. 2488–2492, ISCA.
- [7] Runwu Shi, Benjamin Yen, and Kazuhiro Nakadai, “Distance based single-channel target speech extraction,” in *ICASSP 2025 – 2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. Apr. 2025, pp. 1–5, IEEE.
- [8] Runwu Shi, Zirui Lin, Benjamin Yen, Jiang Wang, Ragib Amin Nihal, and Kazuhiro Nakadai, “Single-channel target speech extraction utilizing distance and room clues,” in *2025 33rd European Signal Processing Conference (EUSIPCO)*. Sept. 2025, pp. 481–485, IEEE.
- [9] Rongzhi Gu and Yi Luo, “Rezero: Region-customizable sound extraction,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 32, pp. 2576–2589, 2024.
- [10] Tuochao Chen, Malek Itani, Sefik Emre Eskimez, Takuya Yoshioka, and Shyamnath Gollakota, “Hearable devices with sound bubbles,” *Nature Electronics*, vol. 7, no. 11, pp. 1047–1058, Nov. 2024.
- [11] Yiheng Jiang, Haoxu Wang, Yafeng Chen, Gang Qiao, and Biao Tian, “Exploring efficient directional and distance cues for regional speech separation,” in *Interspeech 2025*. Aug. 2025, pp. 1478–1482, ISCA.
- [12] Zhong-Qiu Wang, Samuele Cornell, Shukjae Choi, Younglo Lee, Byeong-Yeol Kim, and Shinji Watanabe, “TF-GridNet: Integrating full- and sub-band modeling for speech separation,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 31, pp. 3221–3236, 2023.
- [13] Ethan Perez, Florian Strub, Harm de Vries, Vincent Dumoulin, and Aaron Courville, “FiLM: Visual reasoning with a general conditioning layer,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2018, vol. 32, pp. 3942–3951.
- [14] David Diaz-Guerra, Antonio Miguel, and Jose R. Beltran, “gpuRIR: A python library for room impulse response simulation with GPU acceleration,” in *2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. May 2020, pp. 331–335, IEEE.
- [15] Jonathan Le Roux, Scott Wisdom, Hakan Erdogan, and John R. Hershey, “SDR – half-baked or well done?,” in *2019 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. May 2019, pp. 626–630, IEEE.